Capacity Expansion Network Optimization Report

Model Type: **capacity_expansion**

Metric	Value
Status	Optimal
Objective Value	$730.00
Total Revenue	$790.00
Total Expansion Cost	$60.00
Total Installation Cost	$0.00
Total Cost	$60.00
Net Profit	$730.00
Accepted Demands	5 / 8
Rejected Demands	3 / 8
Acceptance Rate	62.5%
Nodes	7
Edges	10
Total Initial Capacity	430.00
Total Expansion	5.00
Capacity Increase (%)	1.2%

Insights & Highlights

Key Insight	Detail
Most Congested Edge	A -> C (100.0%)
Number of Highly Utilized Edges (≥90%)	6
Most Capacity Expansion	('A', 'C')
Total Lost Revenue	$570.00
Number of Expanded Edges	1

WARNING: Congestion risk in the following edges (utilization ≥90%):
- Edge A -> C: 100.0% used
- Edge B -> D: 100.0% used
- Edge C -> D: 100.0% used
- Edge C -> F: 100.0% used
- Edge D -> E: 100.0% used
- Edge D -> G: 100.0% used

Accepted Demands & Flow Paths

Demand 0: A -> G

Bandwidth: 40.0 units **Revenue:** $200.0
Revenue/BW: $5.00/unit

Path	Fraction	Bandwidth
A -> B -> E -> G	100.00%	40.00

Demand 2: B -> G

Bandwidth: 30.0 units **Revenue:** $150.0
Revenue/BW: $5.00/unit

Path	Fraction	Bandwidth
B -> D -> G	100.00%	30.00

Demand 3: A -> F

Bandwidth: 45.0 units **Revenue:** $220.0
Revenue/BW: $4.89/unit

Path	Fraction	Bandwidth
A -> C -> F	100.00%	45.00

Demand 4: C -> G

Bandwidth: 25.0 units **Revenue:** $120.0
Revenue/BW: $4.80/unit

Path	Fraction	Bandwidth
C -> D -> G	40.00%	10.00
C -> F -> G	40.00%	10.00
C -> D -> E -> G	20.00%	5.00

Demand 6: C -> E

Bandwidth: 20.0 units **Revenue:** $100.0
Revenue/BW: $5.00/unit

Path	Fraction	Bandwidth
C -> D -> E	100.00%	20.00

Rejected Demands

Idx	Source	Target	Bandwidth	Revenue ($)
1	A	E	35.0	180.0
5	B	F	50.0	250.0
7	A	D	30.0	140.0

Total Lost Revenue: $570.00

Capacity Expansions

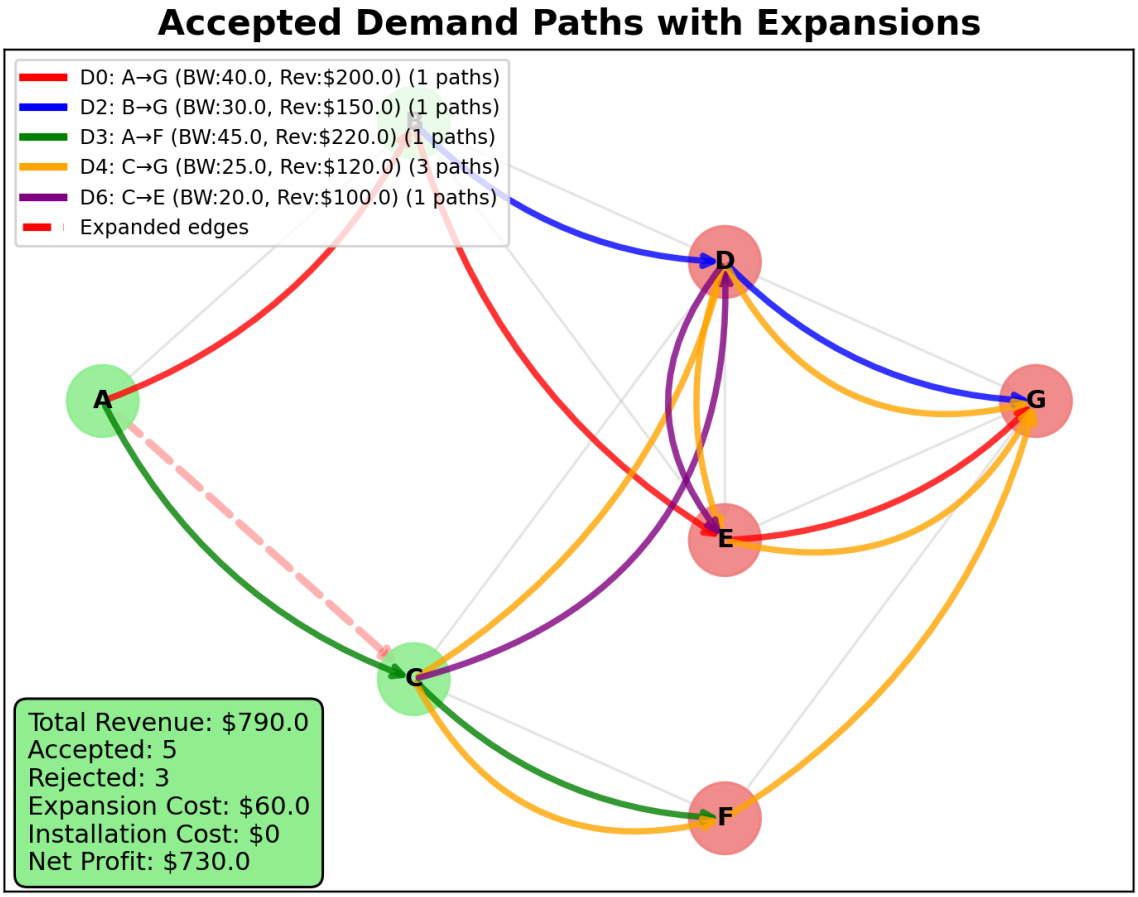
Edge	Expansion (units)	New Capacity	Total Cost ($)
A -> C	5.00	45.00	$60.00

Edge Utilizations (Top 12 Edges)

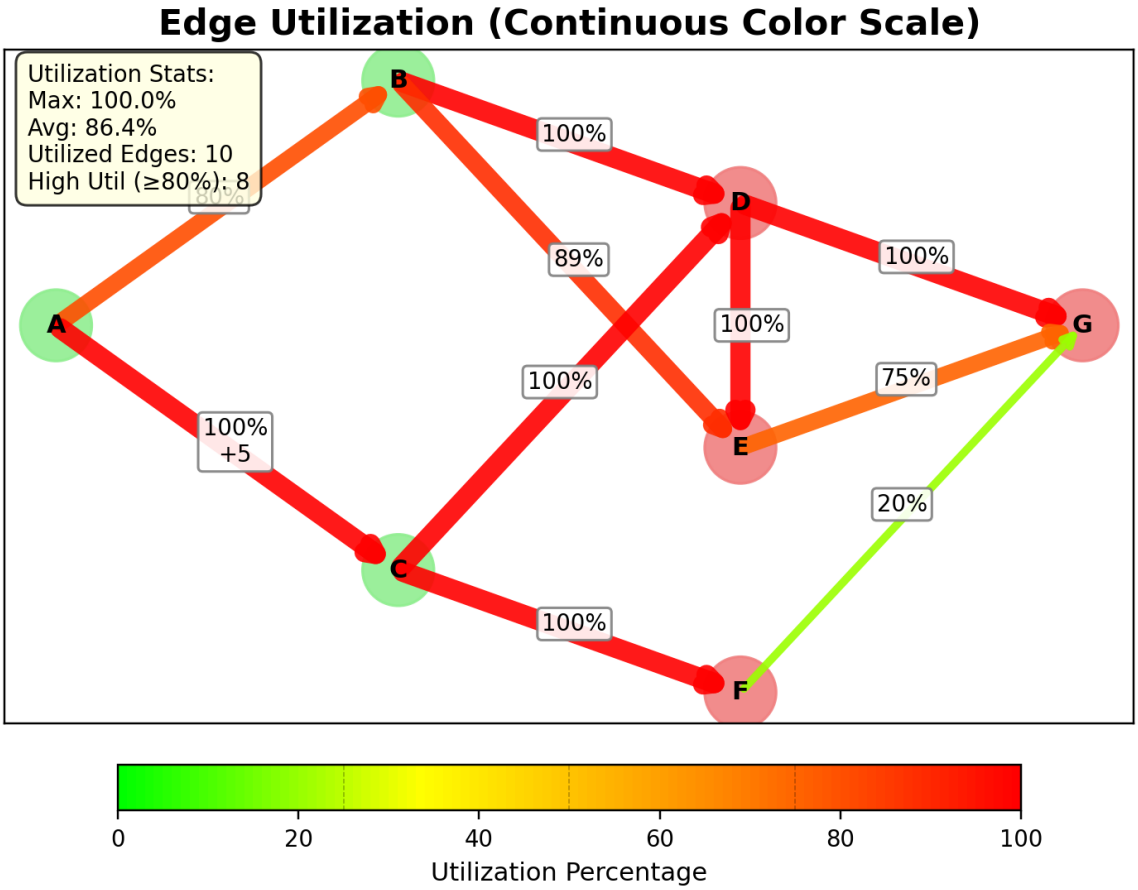
Edge	Flow	Total Capacity	Utilization (%)
A -> C	45.00	45.00	100.0 %
B -> D	30.00	30.00	100.0 %
C -> D	35.00	35.00	100.0 %
C -> F	55.00	55.00	100.0 %
D -> E	25.00	25.00	100.0 %
D -> G	40.00	40.00	100.0 %
B -> E	40.00	45.00	88.9 %
A -> B	40.00	50.00	80.0 %
E -> G	45.00	60.00	75.0 %
F -> G	10.00	50.00	20.0 %

Network Analysis Visualizations

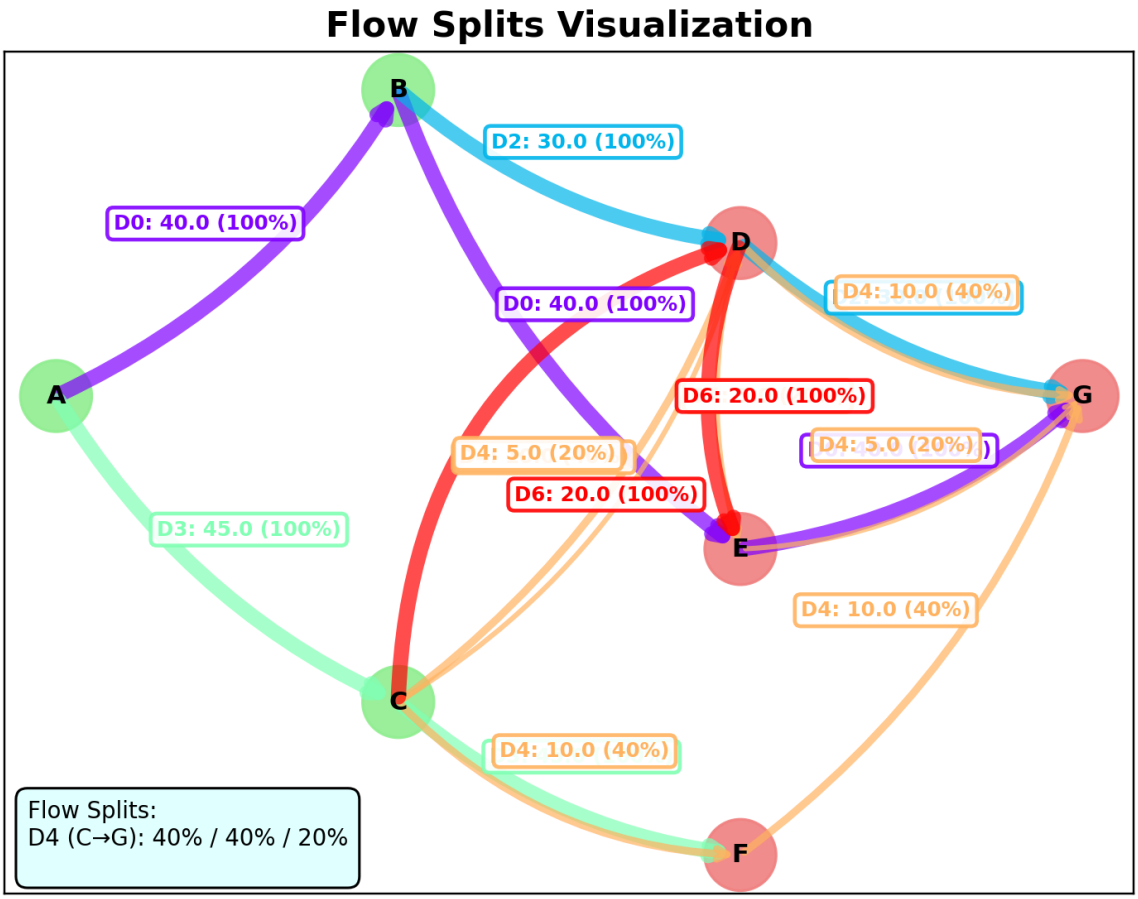
Accepted Demand Paths



Edge Utilization



Flow Splits


Flow Splits Visualization
B
D2: 30.0 (100%)
D0: 40.0 (100%)
D
D4: 10.0 (40%)
D0: 40.0 (100%)
A
D6: 20.0 (100%)
G
D4: 5.0 (20%)
D4: 5.0 (20%)
D6: 20.0 (100%)
D3: 45.0 (100%)
E
D4: 10.0 (40%)
C
D4: 10.0 (40%)
Flow Splits:
D4 (C→G): 40% / 40% / 20%
F

Capacity Expansion Details

